

Fifth Biweekly Project Progress Report (Final Sprint)

Project: PPT Outline Intelligent Generation and Content Completion System

Reporting Period (Sprint 5 — Final): 2026-06-01 – 2026-06-14

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Overall Progress Summary

Sprint 5 is the project's delivery-and-closure sprint, aligned with the Updated Project Charter's final Gantt phases "Experimental Validation & Analysis (05-18 → 06-01)" and "Final Delivery (→ 06-14)". Building on the reliability-hardened prototype from Sprint 4, this sprint took every research and engineering track across the finish line: the live generation path was put through final end-to-end regression, all three evaluation tracks were executed on real APIs (no longer bundled/mock data), and the results were consolidated into a single Technical Validation Report. The project is delivered on the firmly scheduled final date of June 14, 2026.

Headline outcomes this sprint:

- **Final integration regression & frontend–backend fixes (WBS 3.3).** The full stack was regression-tested end to end; remaining frontend–backend interaction defects were diagnosed and closed, stabilizing the "outline confirmation → batch slide generation → streaming display" flow.
- **Multi-model benchmark executed on real APIs (RBS-1 / WBS 2.2.3).** 9 real-API evaluation rounds across Qwen, GLM, and DeepSeek on 3 outline-generation samples were completed; all three models held > 98% schema compliance, and a production model-selection recommendation was delivered.
- **Three-way RAG / DeepResearch comparison executed on real APIs (RBS-3 / WBS 3.1, 3.2).** Pure LLM vs. LLM+RAG vs. LLM+RAG+DeepResearch were scored across two high-difficulty professional domains, combining golden-fact checklist regex matching, ROUGE-L, BERTScore, custom PPT rules, and an LLM-as-Judge factual-accuracy score.
- **Blind human-evaluation track completed (RBS-3 / WBS 3.x).** The full blind-evaluation orchestration was run with two real expert reviewers, replacing the Sprint-4 placeholder scores.
- **Final deliverables packaged.** The prototype deployment was packaged (Dockerized dependencies + STARTUP.md), and the consolidated Technical

Validation Report and Likert Human-Evaluation Criteria were finalized as project deliverables.

The full four-member team is represented in this final report. Total hours invested this sprint: approximately 45 hours. The project is complete and delivered on schedule.

Key Achievements This Sprint

Structured Outline Generation (RBS-1):

The multi-model horizontal comparison (Qwen qwen-plus / GLM glm-4-flash / DeepSeek deepseek-chat) was executed end-to-end on real API calls over 3 outline samples $\times$ 3 runs each. All three models achieved schema-compliance $\geq 98.7\%$, with DeepSeek selected as the primary outline-generation model on the strength of 100% schema compliance and the lowest end-to-end latency (14.64 s avg). The four-level JSON-Schema-constrained outline (chapters 3–6, slides 2–5/chapter, must_cover ≤ 30 chars, slide_intent / takeaway) held across all models.

Knowledge Retrieval & Content Completion (RBS-2):

The content-generation link received its final stability hardening (timeout control across the prompt-generation and retrieval stages), and the batched pgvector RAG retrieval + tightened citation-style handling from Sprint 4 were validated under the live three-way evaluation. RAG reliably grounds static factual data (tariff rates, legal deadlines) into outlines; DeepResearch (Tavily) adds time-sensitive developments (factory timelines, recent rulings).

End-to-End Evaluation System (RBS-3):

All three evaluation tracks were completed and consolidated:

- **Multi-model benchmark** — schema compliance / ROUGE-L / semantic similarity / latency / token cost across Qwen, GLM, DeepSeek.
- **Three-way retrieval comparison** — pure LLM / LLM+RAG / LLM+RAG+DeepResearch on two professional domains, scored by golden-fact regex checklist, ROUGE-L, custom PPT rules, and LLM-as-Judge.
- **Blind human assessment** — Likert-5 rubric (structure, information density, factual accuracy, presentation readiness, overall satisfaction) with two real reviewers and inter-rater reliability analysis (max single-item absolute deviation 0.5).

All results were folded into the unified Technical Validation Report.

Integration & Reliability (WBS 3.3 / 3.4):

Final full-stack regression, frontend–backend interaction fixes, prototype deployment packaging, and the technical-documentation deliverable were completed.

Quantitative Analysis

Per-Member Hours by WBS Task — Sprint 5

WBS ID	Task	Hrs	Status	Compl. %
3.3	Final end-to-end regression + frontend–backend interaction fixes	5	Completed	100%
3.4	Prototype deployment packaging + technical documentation delivery	4	Completed	100%
—	Technical Validation Report consolidation + Milestone 2 review + coordination	3	Completed	100%
3.1/3.2	Execute three-way automated evaluation on real API (LLM / +RAG / +DeepResearch)	8	Completed	100%
3.3	Final stability / timeout hardening of the content-generation link	3	Completed	100%
2.2.3	Multi-model real-API benchmark (Qwen / GLM / DeepSeek, 9 runs) + model-selection recommendation	6	Completed	100%
3.x	Blind human-evaluation orchestration + inter-rater reliability analysis	5	Completed	100%
3.1	Automated three-way evaluation framework + follow-up verification (BERTScore / latency / token)	6	Completed	100%
RBS-3	Consolidate automated + human-eval results into the technical verification report	5	Completed	95%

Group Total Hours per WBS Section (cumulative, all sprints)

WBS Section	Budget (hrs)	Cumulative Hrs Spent	Spent % of Budget	Task Compl. %	Schedule Variance
1. Project Planning & Design	40	40	100%	100%	On schedule
2. Core Module Development	130	~138	~106%	100%	Complete
3. Integration, Evaluation & Delivery	50	~80	~160%	100%	Delivered on 06-14
Total	220	~258	~117%	100%	Delivered on schedule

Final Validation Results (headline)

Multi-model outline benchmark (real API, 3 samples × 3 runs):

Model	Schema Compliance	ROUGE-L	Semantic Sim.	Avg Latency	Cost / call
Qwen (qwen-plus)	98.7%	0.1441	0.8373	34.03 s	¥4.870
GLM (glm-4-flash)	100.0%	0.1957	0.8415	40.13 s	¥0.256
DeepSeek (deepseek-chat)	100.0%	0.1891	0.8812	14.64 s	¥4.539

DeepSeek recommended as the primary online outline-generation model (100% schema compliance, lowest latency); GLM as a low-cost asynchronous-batch candidate; Qwen as a backup.

Three-way retrieval comparison (two-domain average):

Dimension	Pure LLM (B)	LLM+RAG (A)	LLM+RAG+DeepResearch (C)
PPT-rule compliance	95.7%	100.0%	100.0%
ROUGE-L	0.2531	0.2758	0.3198
Golden-fact match rate	25.0%	81.2%	75.0%
Factual accuracy (LLM-Judge)	42.5	85.0	85.0
Blind human composite (1–5)	4.07	4.31	4.52

Key finding: RAG reliably injects static factual data (tariff rates, legal deadlines); DeepResearch adds time-sensitive developments (2025/2026 factory timelines) and lifted the NEV-tariff factual score from 15 → 100, but introduced a date error (2027 vs. 2029) on the Company-Law topic where web sources contained outdated/unofficial content — confirming that DeepResearch needs authoritative-source restriction (e.g. .gov.cn) or RAG cross-checking in high-precision legal/regulatory domains.

Risk Analysis & Management (final status)

#	Risk	L	I	Mitigation / Action	Status
R1	Multi-model benchmark execution slipping past the validation window	Med	Med	Harness reused; 9 real-API runs executed; model-selection recommendation delivered	Closed
R2	Evaluation-phase compression (automated + human-eval both slip)	Low	Med	All three tracks run in parallel on the shared harness; both completed in-sprint	Closed
R3	External data-source instability (Tavily / DashScope rate limits, latency)	Med	Med	Friendly error mapping + bounded retrieval timeout with graceful degradation; per-batch embedding timeout	Mitigated (by design)
R4	Single-developer concentration of core/integration knowledge (bus factor)	Med	Med	STARTUP.md deployment guide; documented module boundaries; reproducible Dockerized bring-up	Managed
R5	Reduced reporting headcount concentrating delivery workload on the team leader	Med	Med	Fourth member (Xie Jingcheng) rejoined for evaluation; core implementation consolidated under team leader	Closed
R6	User trust / acceptance variance of AI-generated content across domains	Low–Med	Med	Citation markers + source-type labelling; stricter citation rules; human-eval + LLM-Judge surfaced and documented the DeepResearch domain gap	Monitoring (documented)

Overall risk posture: the realized high-impact technical risk of prior sprints (content-completion deadlock) remains closed, and every open schedule risk has been closed by completing all evaluation tracks within the sprint. The one residual finding — DeepResearch's reliability gap in strict legal/regulatory domains — is fully documented in the Technical Validation Report with a concrete mitigation for future work.

Ma Xiaolong (2353814 — Team Leader)

This-sprint hours: 12 Work completed (WBS 3.3, 3.4 — final integration, delivery & coordination):

- **Final end-to-end regression & frontend–backend fixes.** Ran a full-stack regression across user frontend ↔ backend ↔ async workers ↔ admin frontend, then diagnosed and closed the remaining frontend–backend interaction defects — stabilizing the complete "outline confirmation → batch slide generation → streaming display" path so it runs cleanly under live API conditions.
- **Prototype deployment packaging & technical documentation.** Packaged the prototype for reproducible bring-up (Dockerized PostgreSQL/pgvector, Redis, MinIO dependencies) and finalized the deployment/technical-documentation deliverable (STARTUP.md).
- **Technical Validation Report consolidation.** Led consolidation of the three evaluation tracks (multi-model benchmark, three-way RAG/DeepResearch comparison, blind human assessment) into the unified Technical Validation Report; curated the report structure and the production model-selection recommendation.
- **Closure & coordination.** Led the Milestone 2 review meeting and final-delivery coordination across the four-member reporting team.

Status: Project delivered on 2026-06-14. All core modules complete; all evaluation tracks consolidated.

Tang Chengyang (2350434)

This-sprint hours: 11 Work completed (WBS 3.1, 3.2, 3.3 — evaluation execution & final stability):

- **Final stability hardening of the content-generation link.** Closed out the stuck/timeout issues at the slide content-completion stage and refined timeout control across the prompt-generation and retrieval stages (task_worker, LLM client, RAG vector retrieval, task controller), making batch slide generation and streaming display robust under load.
- **Three-way comparison executed on real APIs.** Ran the automated evaluation framework — ROUGE, BERTScore, schema compliance, custom PPT rules (must_cover, citation consistency), and factual-accuracy metrics — across the three pathways (pure LLM / LLM+RAG / LLM+RAG+DeepSearch) on two test cases (Transformer NLP, photovoltaic energy), producing the comparison report.
- **Findings.** Confirmed RAG's factual-accuracy improvement and DeepSearch's value-add on web/real-time information.
- **Documentation.** Submitted and updated the third- and fourth-week project progress reports.

Status: Evaluation framework executed and reported; content-generation link finalized.

Zhang Ruiqi (2352017)

This-sprint hours: 11 Work completed (WBS 2.2.3, 3.x — benchmarking & blind evaluation):

- **Multi-model benchmarking.** Ran 9 real-API evaluations across Qwen, GLM, and DeepSeek on 3 outline-generation samples; all three models achieved > 98% schema compliance. Delivered the production model-selection recommendation.
- **Retrieval enhancement pipeline.** Built the three-way comparison framework (LLM-only → LLM+RAG → LLM+RAG+DeepResearch) across two professional domains (EU electric-vehicle tariff policy; China's new Company Law). The automated evaluator combines golden-fact checklist regex matching, ROUGE-L semantic similarity, and LLM-as-Judge factual-accuracy scoring. Key finding: RAG reliably injects static factual data (tariff rates, legal deadlines), while DeepResearch adds time-sensitive developments (factory timelines, recent rulings) but can introduce errors when web sources are outdated or unofficial.
- **Blind evaluation framework.** Developed the full blind-evaluation orchestration — outline shuffling with code-name mapping, a 5-dimension Likert-5 rubric (structure, information density, factual accuracy, presentation readiness, overall satisfaction), and inter-rater reliability analysis. Real expert reviewer scores were collected and folded into the report.

Status: Benchmark, retrieval-comparison, and blind-evaluation results delivered into the Technical Validation Report.

Xie Jingcheng (2252309)

This-sprint hours: 11 Work completed (WBS 3.1, RBS-3 — evaluation framework & verification):

- **Systematic benchmark experiments.** Conducted real-API benchmark tests on Qwen, GLM, and DeepSeek — 9 evaluation rounds over 3 outline samples — verifying all three models held > 98% schema compliance.
- **Three-way comparative framework.** Built the automated evaluation system integrating regex-based fact checking, ROUGE-L semantic similarity, and LLM-as-Judge accuracy scoring across LLM-only / LLM+RAG / LLM+RAG+DeepResearch, validated on two professional domains (EU EV tariff policy; China's revised Company Law).
- **Follow-up verification.** Ran repeated comparative tests with unified prompts and JSON schemas, recording schema compliance, ROUGE-L, BERTScore, response latency, and token cost; developed customized automated evaluation scripts to quantitatively compare the three technical solutions, confirming RAG's factual-accuracy gain and DeepResearch's domain-specific value-add (and its outdated-source risk).
- **Report consolidation.** Developed the blind-evaluation pipeline (outline shuffling, code-name mapping, Likert-5 rubric, inter-rater reliability) and integrated automated + human-assessment results into the comprehensive technical verification report.

Status: Evaluation framework and verification complete; results consolidated into the Technical Validation Report.